Computational approach to jigsaw puzzles

Friday sparks

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2026/03/20

History of jigsaw puzzles

- Cartographer John Spilsbury believed to have produced the first jigsaw puzzle around 1760 using a marquetry saw;

Modern jigsaw puzzles

- Made from paperboard;
- Worlds largest puzzle had 551,232 pieces and was assembled by 1,600 students in Vietnam;
- Wikipedia's logo is a puzzle;
- From 2019, there's an annual World Jigsaw Puzzle Championship;

Swedish puzzle championship

Swedish puzzle championship

Swedish puzzle championship

Swedish puzzle championship

Types of puzzles

Strategies

Some common strategies

- Flip all pieces right side up;
- Sort out edge pieces;
- Separate into distinguishable colours;
- Use sorting boxes/plates;
- Sort by shape (brute force);
- Start by making the sky (if you're me)

How fast are the speed puzzlers?

- Puzzle from the individual final:

How fast are the speed puzzlers?

Classification Individual

Final

14/03/2026 17:45

Partille Arena - Gamla Kronvägen 2, 433 33 Partille

500 01:15:00

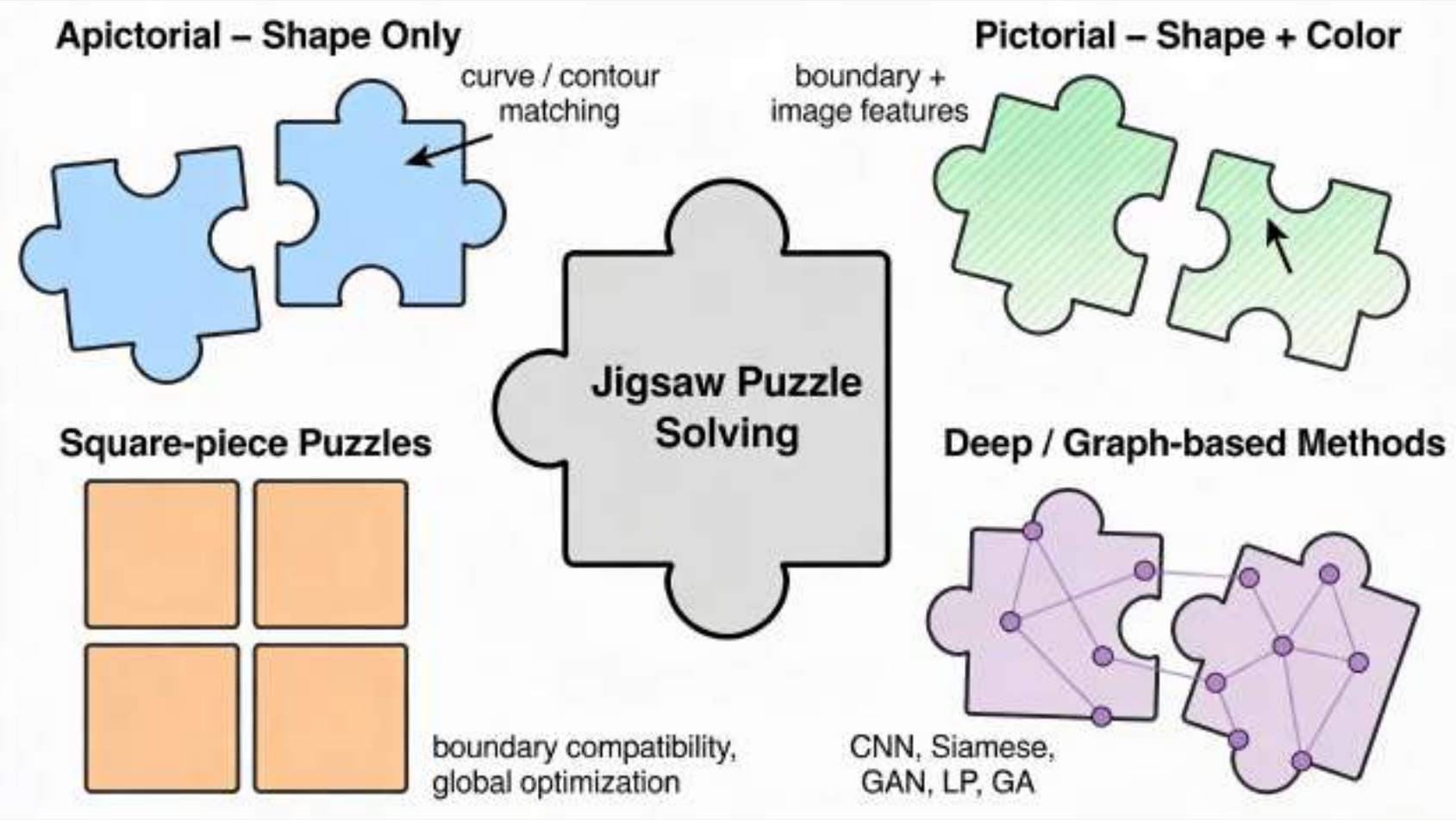
General International National

#	Name	Origin	Country	Time		
1	Kerstin Svensson	Partille (Västra Götaland)	Sweden	00:32:02	-	-
2	Emilia Persson	Ystad (Skåne)	Sweden	00:33:49	+1' 47"	at 1' 47"
3	Bodil Norberg Scherman	Bandhagen (Södermanlan)	Sweden	00:34:06	+ 17"	at 2' 04'
4	Line Beate Svestad	Oslo	Norway	00:34:56	+ 50"	at 2' 54"
5	Tyra Weiring	Västerås (Västmanland)	Sweden	00:34:57	+ 01'	at 2' 55"
6	Annemie Kostezer	Basel (BS)	Switzerland	00:35:56	+ 59"	at 3' 54"
7	Merle Kleemann	Aachen (Nrw)	Germany	00:36:43	+ 47"	at 4' 41"
8	Tobias Priemer	Dalby (Skåne)	Sweden	00:37:21	+ 38"	at 5' 19"

How fast are the speed puzzlers?

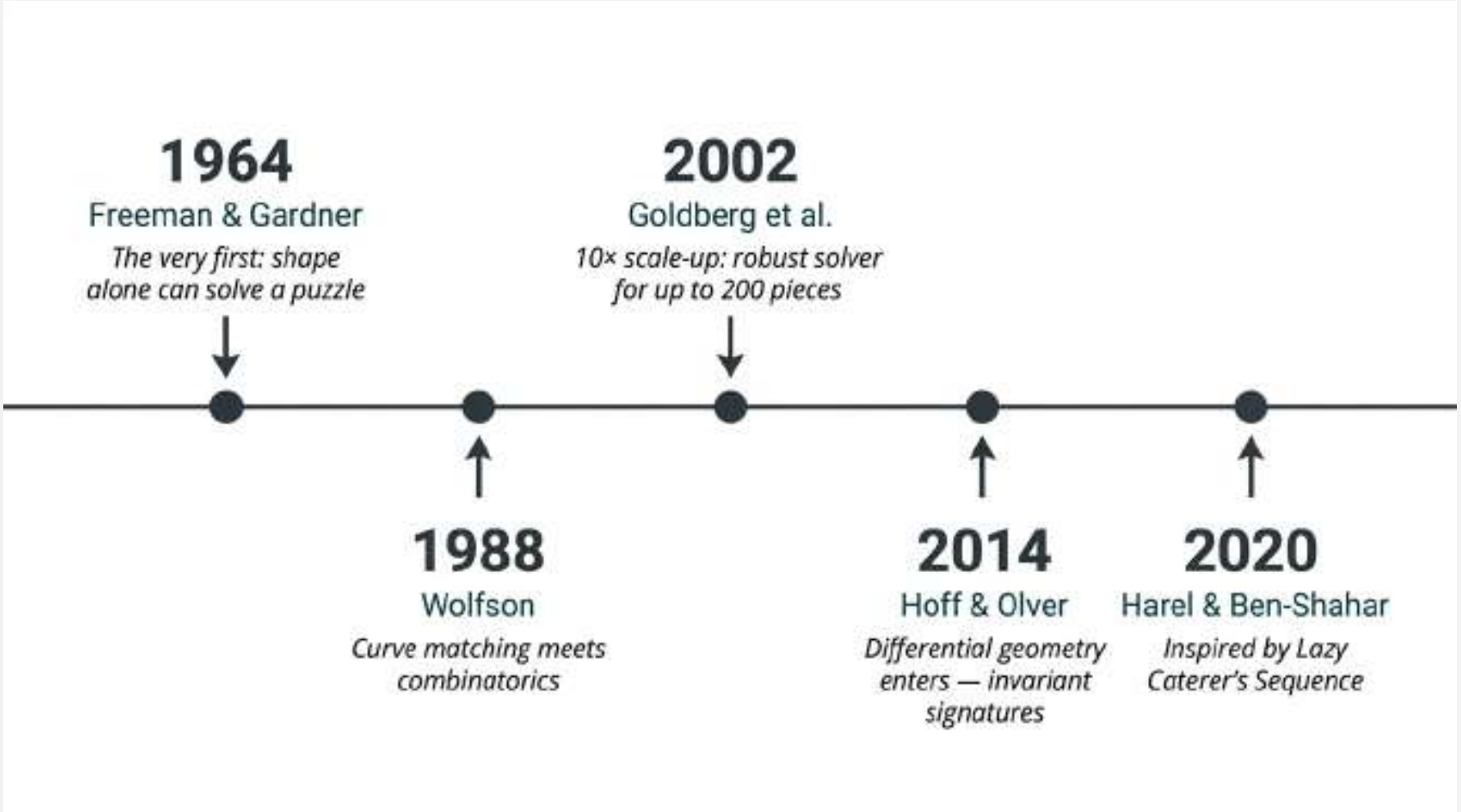
Is there a more mathematical approach to solving puzzles?

Computational approach

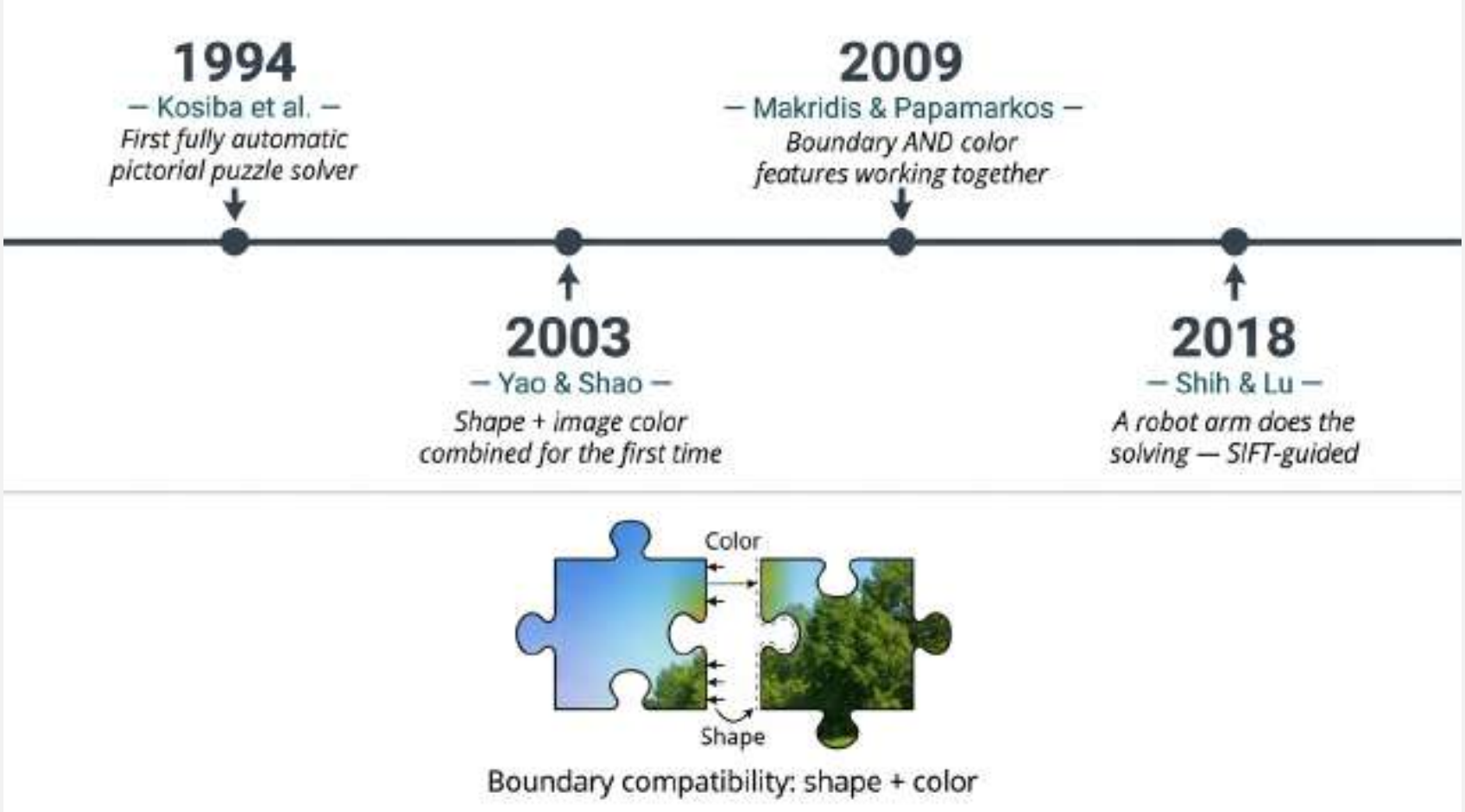


- Markaki S, Panagiotakis C. Jigsaw puzzle solving techniques and applications: a survey. The Visual Computer. 2023 Oct;39(10):4405-21.

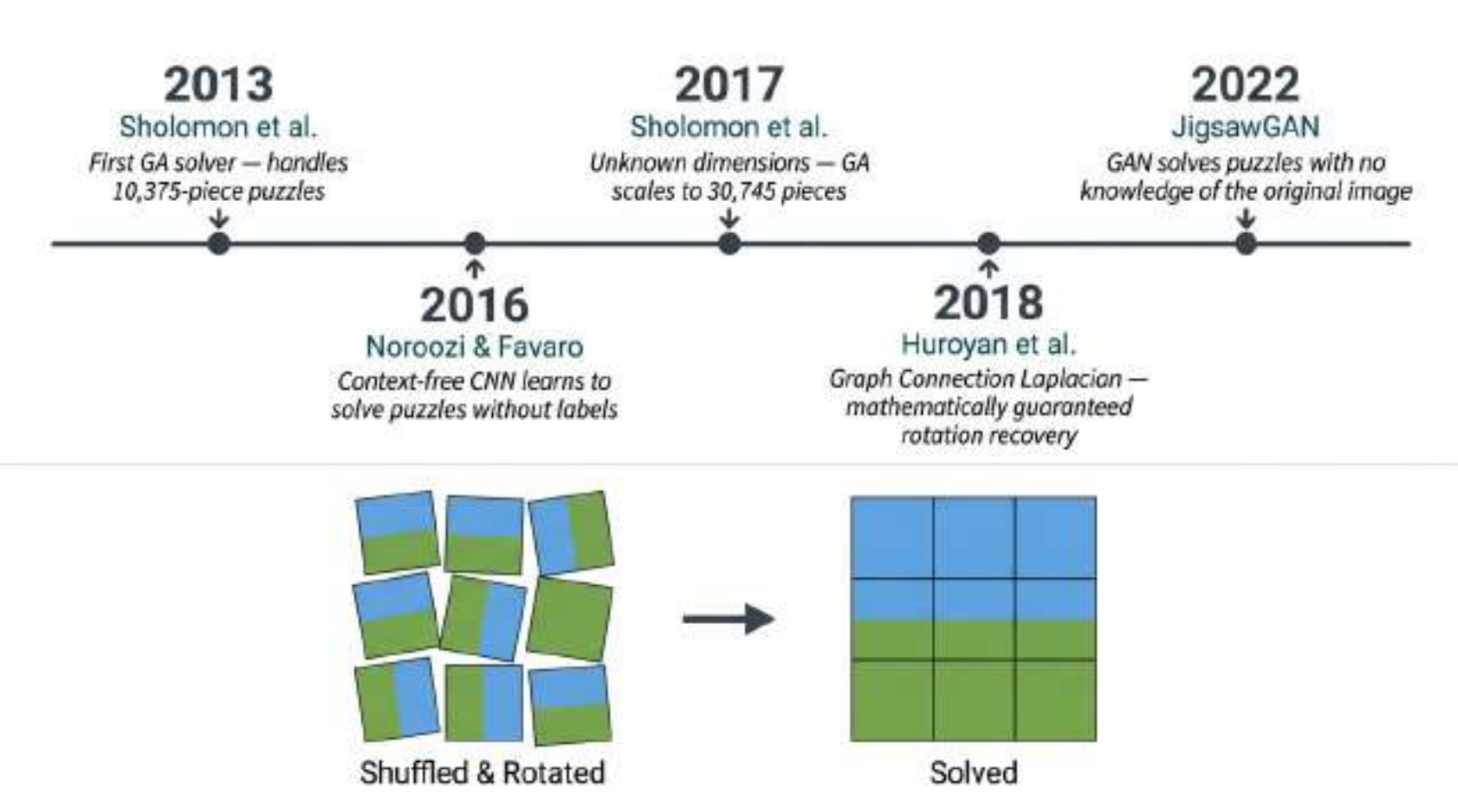
From first solutions of apictorial puzzles



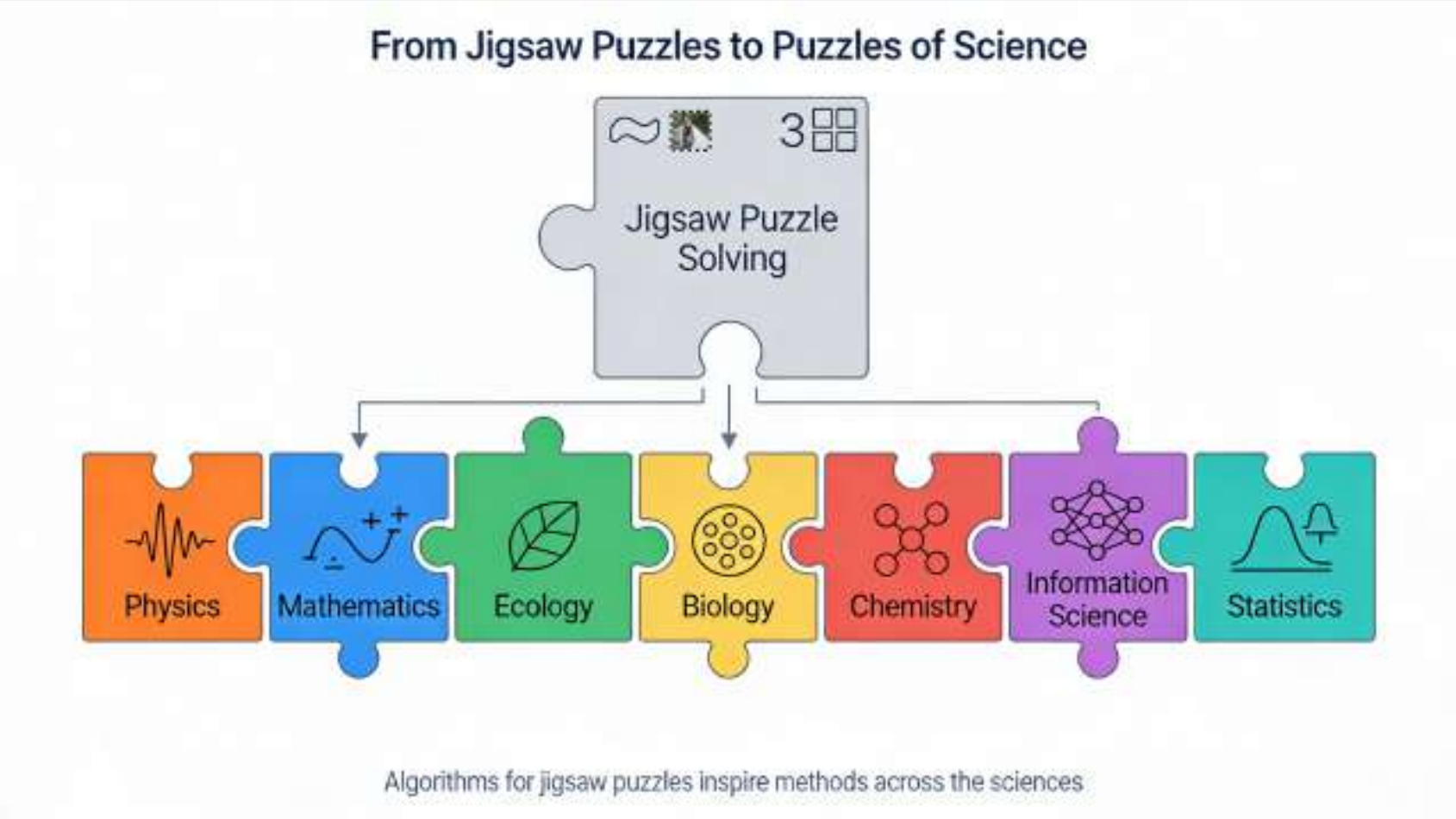
To the pictorial ones



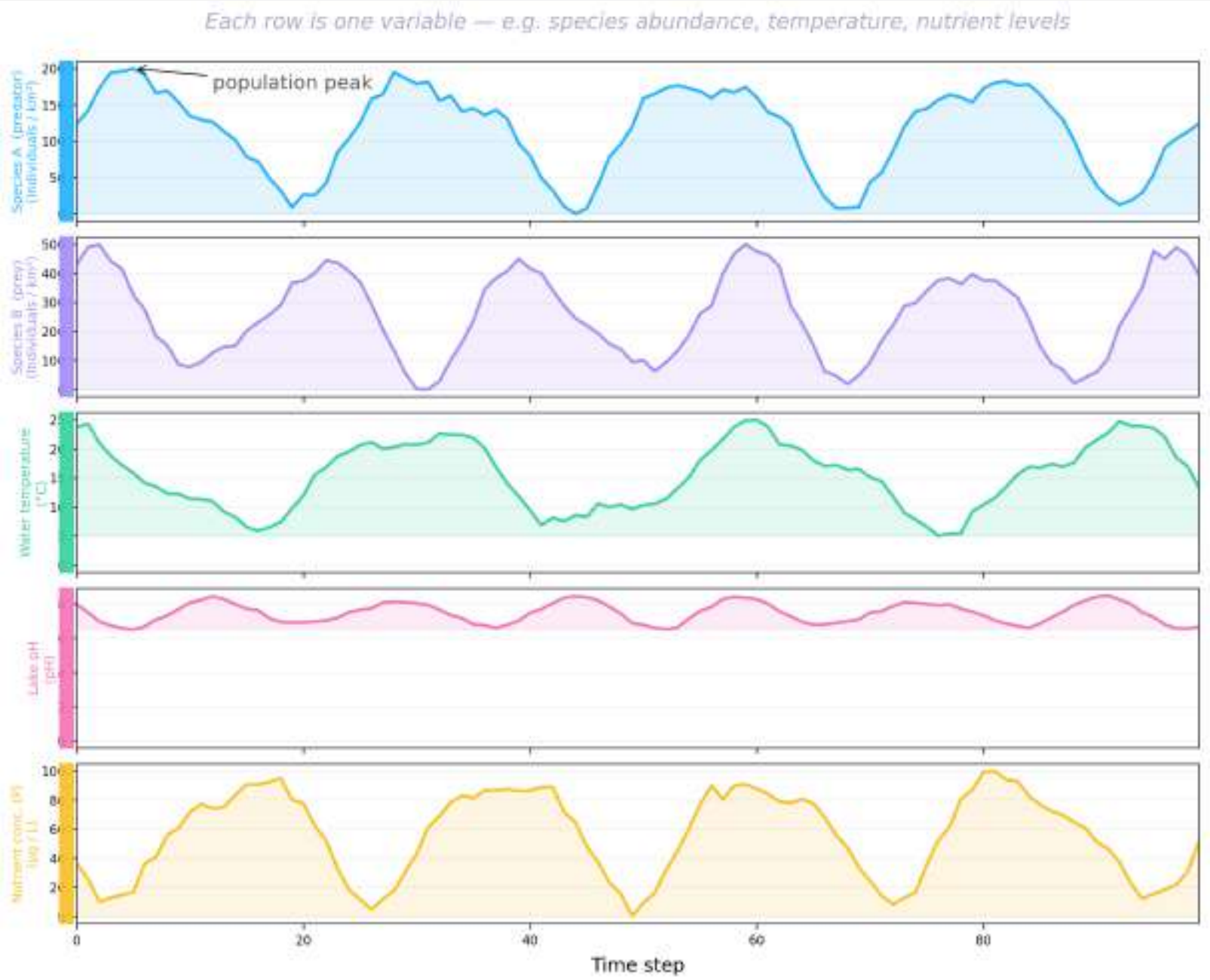
and square puzzles

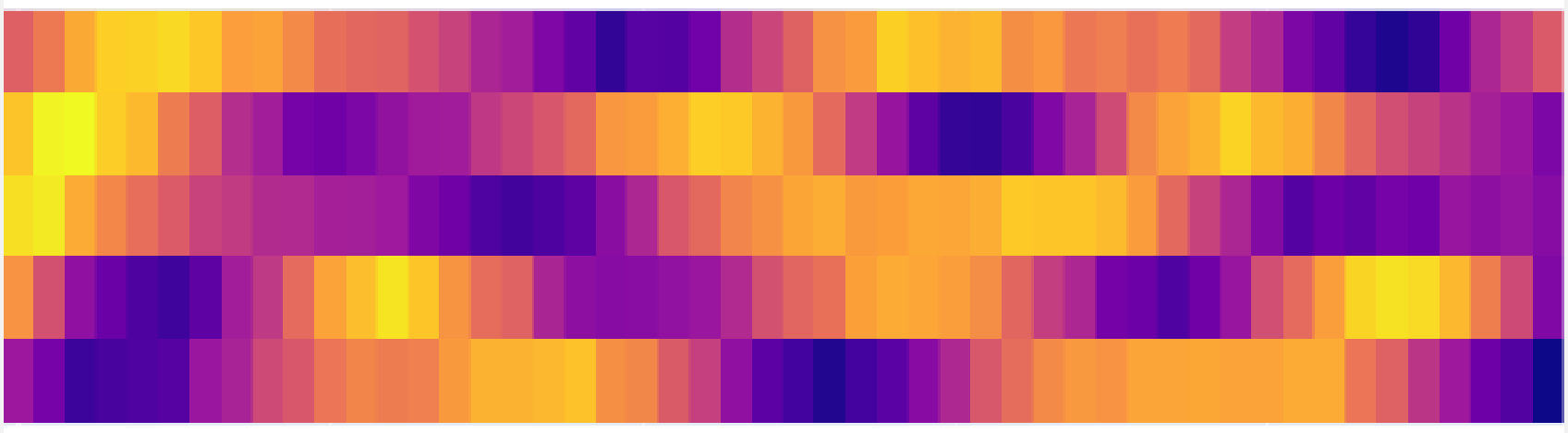


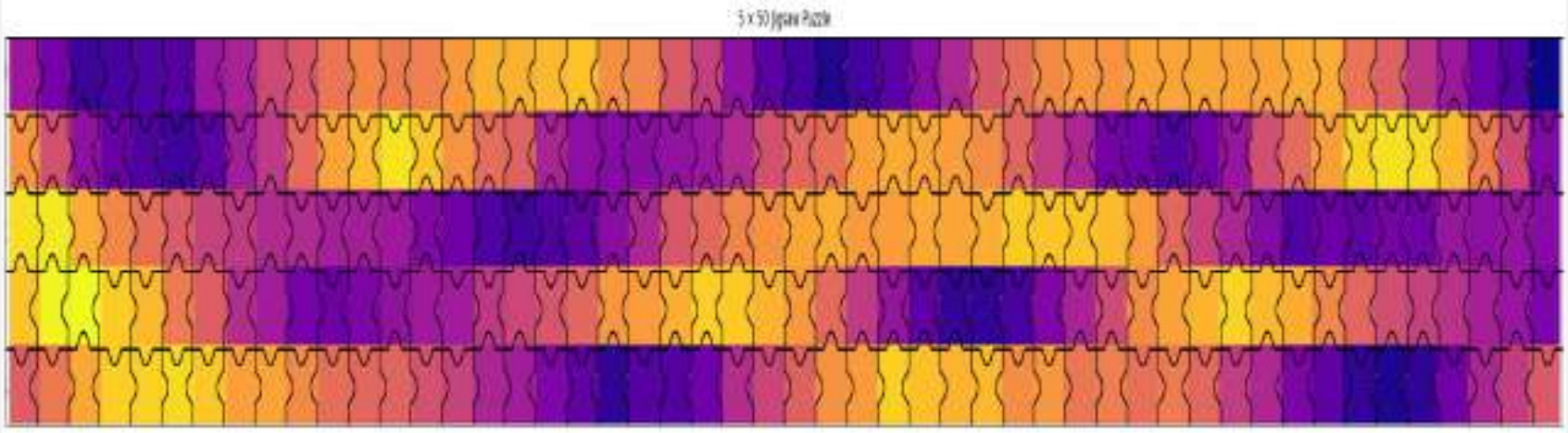
Why would a computer science puzzle algorithm matter to an ecologist or an interdisciplinary?

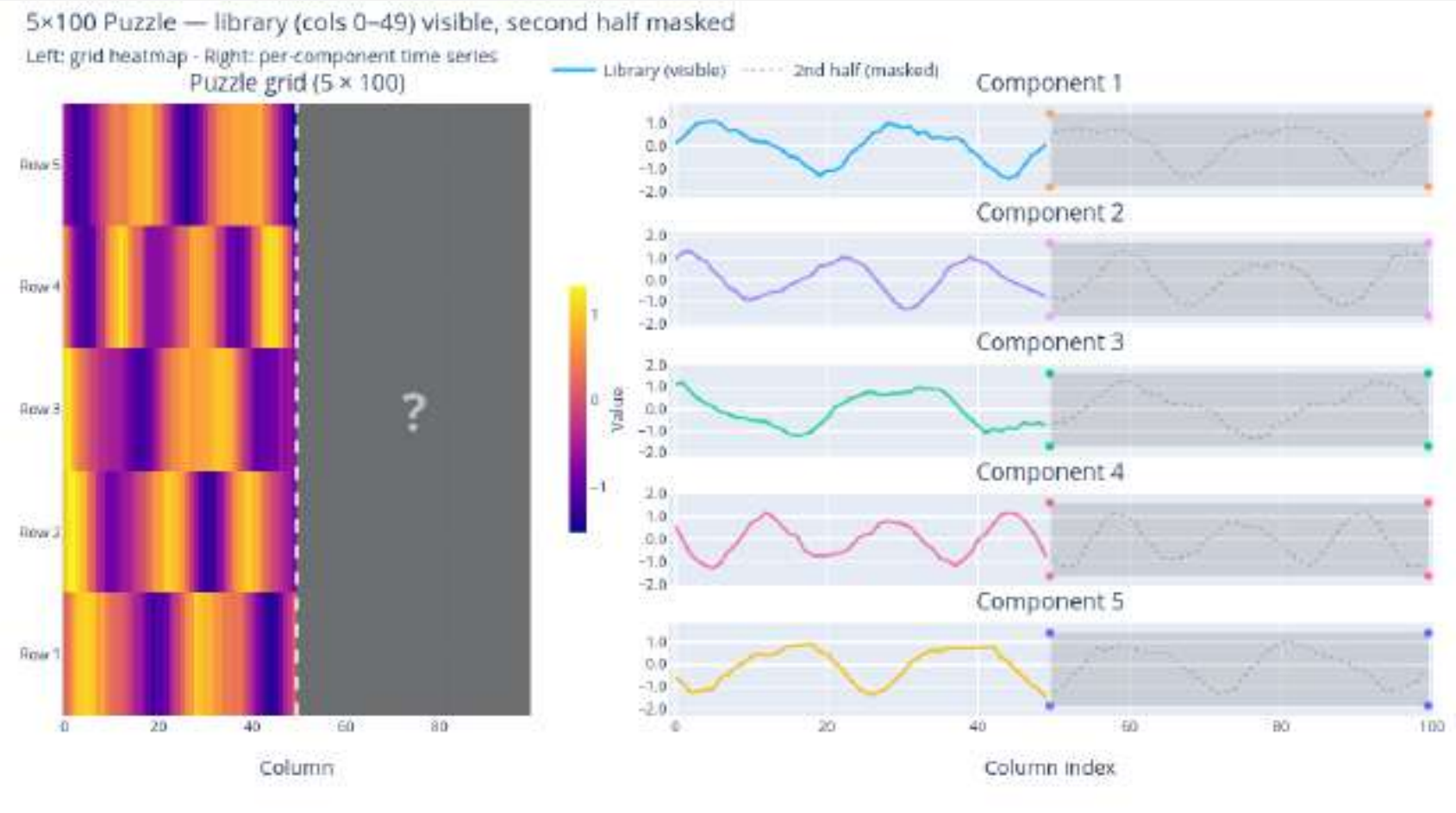

From Jigsaw Puzzles to Puzzles of Science
3
Jigsaw Puzzle Solving
Physics
Mathematics
Ecology
Biology
Chemistry
Information Science
Statistics
Algorithms for jigsaw puzzles inspire methods across the sciences

Can some research problems be "popularized" by interpreting them as puzzles?


Each row is one variable — e.g. species abundance, temperature, nutrient levels
population peak
Species A (predator) (individuals / km²)
Species B (prey) (individuals / km²)
Water temperature (°C)
Lake pH (pH)
Nutrient conc. (P) (µg / L)
0
20
40
60
80
Time step




5 x 50 Jigsaw Puzzle


5×100 Puzzle — library (cols 0–49) visible, second half masked
Left: grid heatmap - Right: per-component time series
Puzzle grid (5 × 100)
Row 5
Row 4
Row 3
Row 2
Row 1
?
Column
Library (visible)
2nd half (masked)
Value
Component 1
Component 2
Component 3
Component 4
Component 5
Column index

Empirical approach

1. **Observation vector.** Each time step t yields a p-dimensional observation:

$$\mathbf{x}(t) = \left[x_1(t),\ x_2(t),\ \ldots,\ x_p(t)\right]^\top \in \mathbb{R}^p$$

2. **Delay embedding (Takens, 1981).** Reconstruct the state space by stacking E lagged copies separated by delay τ:

$$\mathbf{v}(t) = \left[\mathbf{x}(t),\ \mathbf{x}(t-\tau),\ \ldots,\ \mathbf{x}(t-(E-1)\tau)\right]^\top \in \mathbb{R}^{pE}$$

   Parameters E (embedding dimension) and τ (delay) are chosen to best unfold the attractor.

3. **Library construction.** The first half of the time series forms the *library* $\mathcal{L}$ of n embedded state vectors, each paired with its τ-ahead target:

$$\mathcal{L} = \left\{ \left(\mathbf{v}(t_i),\ \mathbf{x}(t_i+\tau)\right) \right\}_{i=1}^{n}$$

Empirical approach 2

1. **Query vector.** Given the last observed columns, build the current state:

$$\mathbf{v}^* = \left[\mathbf{x}(T),\ \mathbf{x}(T-\tau),\ \ldots,\ \mathbf{x}(T-(E-1)\tau)\right]^\top$$

2. **Nearest-neighbour search.** Find the $E+1$ library vectors closest to $\mathbf{v}^*$ in Euclidean distance:

$$d_j = \left\|\mathbf{v}^* - \mathbf{v}(t_j)\right\|_2, \qquad j \in \mathcal{N}_{E+1}(\mathbf{v}^*)$$

3. **Exponential weights (Simplex Projection).** Weight each neighbour by proximity, normalised to sum to one:

$$w_j = \frac{\exp(-d_j \,/\, d_{\min})}{\sum_k \exp(-d_k \,/\, d_{\min})}, \qquad d_{\min} = \min_j d_j$$

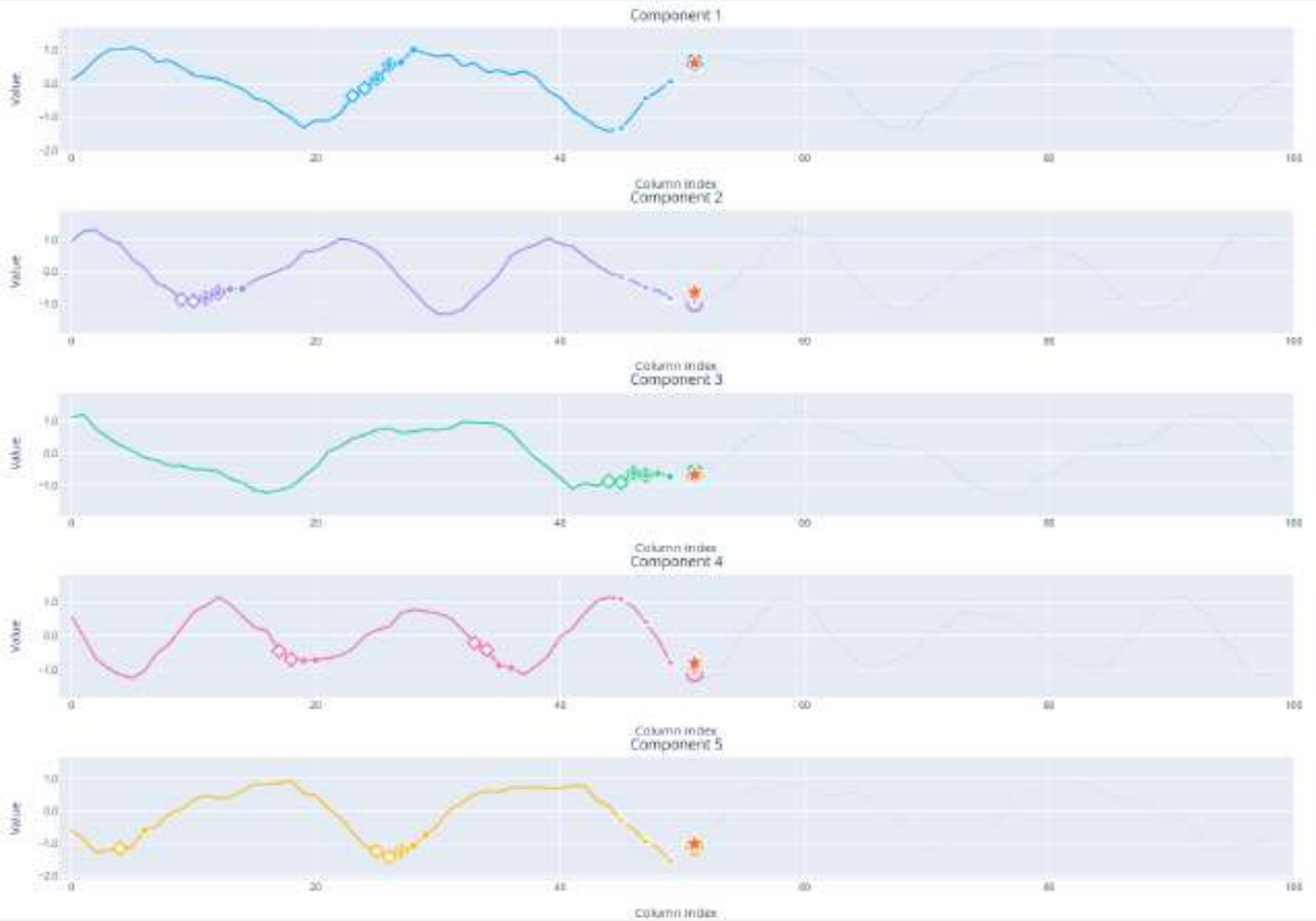
Empirical approach 3

1. **Prediction.** The forecast τ steps ahead is the weighted average of the neighbours' future values:

$$\boxed{\hat{\mathbf{x}}(T+\tau) = \sum_{j \in \mathcal{N}_{E+1}} w_j \cdot \mathbf{x}(t_j + \tau)}$$

2. **Iterative completion.** Append $\hat{\mathbf{x}}(T+\tau)$ to the library and repeat steps 4–7 to reconstruct the full second half of the puzzle, column by column.

Empirical dynamical modelling approach



What other scientific problems at IceLab could be reframed as a jigsaw puzzle?